Quantum Field Theory – Homework 5

Name: ______________________ **Due Date:** 2025.11.24

The following problems are based on Lectures 9 and 10, focusing on QFT propagators, their definitions, causal structure, and Fourier-space representation. The bullet points are guiding hints to help structure your answers.

Problem 1. Equations of Motion for Propagators

In a real scalar quantum field theory, the field operator satisfies the Klein-Gordon equation

$$(\partial^2 + m^2)\,\hat{\phi}(x) = 0.$$

The two Wightman functions are defined as

$$G_+(x,x') = \langle 0|\hat{\phi}(x)\hat{\phi}(x')|0\rangle, \qquad G_-(x,x') = \langle 0|\hat{\phi}(x')\hat{\phi}(x)|0\rangle.$$

The retarded, advanced, and Feynman propagators are given by

$$G_R(x,x') = \theta(t-t')\,\langle 0|[\hat{\phi}(x),\hat{\phi}(x')]|0\rangle, \tag{1}$$

$$G_A(x,x') = -\theta(t'-t)\,\langle 0|[\hat{\phi}(x),\hat{\phi}(x')]|0\rangle, \tag{2}$$

$$G_F(x,x') = \theta(t-t')\,G_+(x,x') + \theta(t'-t)\,G_-(x,x'). \tag{3}$$

(a) Show that the Wightman functions satisfy the homogeneous Klein-Gordon equation:

$$(\partial^2 + m^2)\,G_\pm(x,x') = 0.$$

(b) Compute $(\partial^2 + m^2)\,G_R(x,x')$ and show that the retarded propagator satisfies a Greens function equation with a delta-function source.

(c) Repeat the computation for $G_A(x,x')$ and verify that it satisfies the same Greens function equation.

(d) Show that the Feynman propagator satisfies

$$(\partial^2 + m^2)\,G_F(x-x') = -\,i\,\delta^{(4)}(x-x'),$$

consistent with its Fourier-space representation.

Problem 2. Cauchy Principal Value Integral (20 points)

Consider the integral

$$I(k) = \text{P.V.} \int_{-\infty}^{\infty} \frac{e^{ikx}}{x}\, dx,$$

where k is a real constant and P.V. denotes the Cauchy principal value.

In this problem we focus on the case $k < 0$ and evaluate the integral using complex contour methods.

Problem 3. Advanced Propagator (20 points)

For a real scalar field, the advanced propagator is defined by

$$G_A(x,x') \equiv -\,\theta(t'-t)\,\langle 0|[\hat{\phi}(x),\hat{\phi}(x')]|0\rangle,$$

where

$$[\hat{\phi}(x),\hat{\phi}(x')] = \hat{\phi}(x)\hat{\phi}(x') - \hat{\phi}(x')\hat{\phi}(x)$$

and θ is the step function. The field operator has the mode expansion

$$\hat{\phi}(x) = \int \frac{d^3k}{(2\pi)^3}\,\frac{1}{\sqrt{2\omega_{\vec{k}}}}\left(\hat{a}_{\vec{k}}\,e^{-i\omega_{\vec{k}}t+i\vec{k}\cdot\vec{x}} + \hat{a}^\dagger_{\vec{k}}\,e^{+i\omega_{\vec{k}}t-i\vec{k}\cdot\vec{x}}\right), \qquad \omega_{\vec{k}} = \sqrt{\vec{k}^2+m^2},$$

and the Wightman functions are

$$G_+(x,x') = \langle 0|\hat{\phi}(x)\hat{\phi}(x')|0\rangle, \qquad G_-(x,x') = \langle 0|\hat{\phi}(x')\hat{\phi}(x)|0\rangle.$$

(a) Show that the commutator expectation value can be written as

$$\langle 0|[\hat{\phi}(x),\hat{\phi}(x')]|0\rangle = G_+(x,x') - G_-(x,x').$$

Hence express $G_A(x,x')$ in terms of $G_+(x,x')$ and $G_-(x,x')$.

(b) Using the mode expansion of $\hat{\phi}(x)$ and the vacuum conditions

$$\hat{a}_{\vec{k}}|0\rangle = 0, \qquad \langle 0|\hat{a}^\dagger_{\vec{k}} = 0, \qquad [\hat{a}_{\vec{k}},\hat{a}^\dagger_{\vec{k}'}] = (2\pi)^3\delta^{(3)}(\vec{k}-\vec{k}'),$$

derive the explicit momentum-space expressions

$$G_+(x,x') = \int \frac{d^3k}{(2\pi)^3}\,\frac{1}{2\omega_{\vec{k}}}\,e^{-i\omega_{\vec{k}}(t-t')+i\vec{k}\cdot(\vec{x}-\vec{x}')},$$

$$G_-(x,x') = \int \frac{d^3k}{(2\pi)^3}\,\frac{1}{2\omega_{\vec{k}}}\,e^{+i\omega_{\vec{k}}(t-t')-i\vec{k}\cdot(\vec{x}-\vec{x}')}.$$

(c) Using your results from parts (a) and (b), derive the exact form of the advanced propagator:

$$G_A(x,x') = -\,\theta(t'-t)\int \frac{d^3k}{(2\pi)^3}\,\frac{1}{2\omega_{\vec{k}}}\left[e^{-i\omega_{\vec{k}}(t-t')+i\vec{k}\cdot(\vec{x}-\vec{x}')} - e^{+i\omega_{\vec{k}}(t-t')-i\vec{k}\cdot(\vec{x}-\vec{x}')}\right].$$

(d) Show that $G_A(x,x') = 0$ for $t > t'$ and briefly explain why this is the correct *advanced* causal behavior.

(e) Finally, act with the Klein–Gordon operator on the x-argument and show that

$$(\partial^2 + m^2)\, G_A(x, x') = -\, i\, \delta^{(4)}(x - x'),$$

so that the advanced propagator is a Greens function of the Klein–Gordon operator with advanced boundary condition.